

Programme : B.Tech
Year / Semester : II / II
Branch / Section : IT/CSE
Course Code : 23CD4T02
Subject : Database Management System

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QUESTION BANK FOR MID-II**UNIT -3****Short answer questions**

Q. No.	Question	M	CO	LL
1	Differentiate between Subquery and Correlated Subquery	2	3	LL1
2	List aggregate functions in SQL.	2	3	LL1
3	Differentiate between WHERE and HAVING clause.	2	3	LL1
4	Define joins in SQL.	2	3	LL1
5	Define a View in SQL. Give the syntax for creating a view.	2	3	LL1

Essay questions

Q. No.	Question	M	CO	LL
1	Explain the use of IN, ANY, ALL, and EXISTS operators in nested queries with suitable examples.	6	3	LL2
2	Discuss GROUP BY and HAVING clauses in SQL with suitable examples.	6	3	LL2
3	Explain different types of SQL joins with syntax and examples.	6	3	LL2
4	Explain the operations on a view in SQL with suitable examples.	6	3	LL2
5	Explain relational set operators in SQL with examples	6	3	LL2
6	a) Write a SQL query with a complex WHERE clause to retrieve employees whose salary is between 50000 and 70000 (inclusive), whose last name does not start with 'S', and whose first name contains 'a' from EMPLOYEE TABLE b) Write a SQL query to find the 2nd highest salary from EMPLOYEE table using a subquery.	6	3	LL3

UNIT-4**Short answer questions**

Q. No.	Question	M	CO	LL
1	Define Normalization and state its advantages	2	4	LL1
2	Define Super Key and Candidate Key.	2	4	LL1
3	State Functional Dependency.	2	4	LL1
4	Differentiate between Trivial and Non-Trivial Functional Dependency	2	4	LL1
5	Define Fully Functional Dependency	2	4	LL1
6	Define Partial Functional Dependency	2	4	LL1

7	Define Transitive Functional Dependency	2	4	LL1
8	Define First Normal Form	2	4	LL1
9	Define Multivalued Dependency.	2	4	LL1
10	Define Lossless Join and Dependency Preserving Decomposition	2	4	LL1

Essay questions

Q. No.		Question	M	CO	LL
1	a	Explain the purpose of normalization with suitable examples.	6	4	LL2
	b	Explain various keys in DBMS with examples.	6	4	LL2
2	a	Explain the different types of Functional Dependencies with examples.	6	4	LL2
	b	Discuss the Properties of Functional Dependency.	6	4	LL2
3	a	Explain Second Normal Form (2NF) with a suitable example.	6	4	LL2
	b	Explain Third Normal Form (3NF) with a suitable example.	6	4	LL2
4	a	Explain Boyce–Codd Normal Form (BCNF) with a suitable example.	6	4	LL2
	b	Explain the lossless and dependency preserving decomposition.	6	4	LL2
5	a	Discuss Fourth Normal Form (4NF) with a suitable example.	6	4	LL2
	b	Discuss Fifth Normal Form (5NF) with a suitable example.	6	4	LL2
6	a	Given relation R(A,B,C,D) has the FD {A->B, C->D}. Is this relation in BCNF? Justify your answer with a proper explanation of the BCNF Condition	6	4	LL2
	b	Determine the closer of the following set of functional dependencies for a relation scheme. R(A,B,C,D,E) into R1(A,B,C) and R2(A,D,E) determine that this decomposition is a lossless-join decomposition or dependency preserving if the following set F of FD's holds: A->BC, CD->E, B->D, E->A.	6	4	LL3

UNIT-5

Short answer questions

Q. No.	Question	M	CO	LL
1	Define a Transaction in DBMS.	2	5	LL1
2	Define Atomicity in a transaction.	2	5	LL1
3	Define consistency in a transaction.	2	5	LL1
4	Define Schedule in DBMS.	2	5	LL1
5	Define File Organization and mention its purpose.	2	5	LL1
6	Define Indexing in DBMS. State its purpose.	2	5	LL1
7	Define a clustered index	2	5	LL1
8	Differentiate between sparse index and dense index.	2	5	LL1
9	Differentiate between Primary index, Secondary Index	2	5	LL1
10	Differentiate between Indexing and Hashing.	2	5	LL1

Essay questions

Q. No.		Question	M	CO	LL
1	a	Explain different operations of a transaction in DBMS with suitable examples.	6	5	LL2
	b	Explain Database Buffers in DBMS and its advantages.	6	5	LL2
2	a	Explain different types of failures in transaction processing.	6	5	LL2
	b	Explain Log-based Recovery in DBMS	6	5	LL2

3	a	Explain the various states of a Transaction with a neat diagram	6	5	LL2
	b	Explain the ACID Properties of Transaction with examples	6	5	LL2
4	a	Explain different types of Schedules with examples.	6	5	LL2
	b	Consider the schedule $S=\{R_1(A), W_2(B), W_1(A), C_1, W_2(A), C_2\}$. Analyze whether this schedule is Conflict Serializable. If not, list the conflicting operations that prevent conflict serializability by drawing the precedence graph.	6	5	LL2
5	a	Describe about File Organization in DBMS	6	5	LL2
	b	Explain various Indexing Methods in DBMS	6	5	LL2
6	a	Explain Hash Based Indexing	6	5	LL2
	b	Explain Tree Based Indexing	6	5	LL2

Course Coordinator

Module Coordinator

Programme Coordinator

Coordinator IQAC